# Object Memory under Natural Occlusion: Learned Appearance, Spatial Association, and the Cost of Incorrect Recall

Executed research study and development report

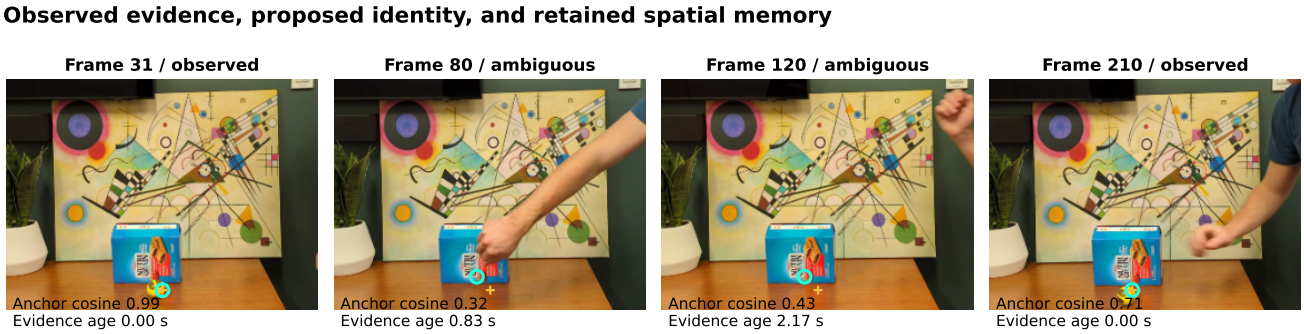



Figure 1. **One recorded episode, four measured states.** A hand and moving box obscure the selected target in TCOW Rubric Office. Cyan circles are the spatial-mixture candidate; amber crosses mark the last supported coordinate. Frame states, anchor similarity and evidence age come from the executed causal trace. An accepted match is not a ground-truth identity assertion. No humanoid drawing substitutes for an executed robot.


## Abstract

*An agent answering questions about an occluded object must distinguish retained evidence from a newly verified identity. We test whether integrating spatial memory into learned visual association improves this distinction. A frozen DINOv2 encoder supplies identical candidate sets to appearance-only, single-motion, multiple-hypothesis, and controlled-memory-update variants. Unlike the earlier classical prototype, spatial hypotheses directly change ranking: the mixture changes 1,934 of 7,530 frame selections. We execute all four variants on 36 clips and 56 sparse target-mask endpoints across two TCOW collections. Learned appearance yields 22 candidate hits out of 31 on Office, versus nine for the earlier classical proposal; the protocols differ in representation and query use. However, spatial association does not improve accepted hits and increases accepted misses in both collections. We release all outcomes, six outcome-stratified examples, source-verified references, and a grounded response interface. An official Ameca Animator project was created, but responsive speech could not be executed because voices and the selected script runtime were unavailable. This report establishes implementation and a failure diagnosis, not a validated novel occlusion resolver, calibrated hidden localization, or humanoid transfer.*


## 1. Problem and evidence contract

A humanoid may be asked which object was placed behind a box, where it was last supported by visual evidence, or whether a similar object that reappeared is the original. Four capabilities are distinct: retaining identity evidence, propagating spatial alternatives, verifying identity at return, and deciding whether to answer or acquire another observation. A system can preserve an RGB crop while failing at all three later capabilities.

Our motivating failure is incorrect recall with unwarranted certainty. A tracker may follow a foreground hand or accept a similar cup and then overwrite the evidence that would have revealed its error. Conversely, a conservative system can appear precise by rejecting almost everything. Useful continuity requires both accepted correct associations and controlled false associations. We therefore report hits, misses and abstentions together.

The operational task here is narrower than that motivation. A query mask initializes one identity representation. The algorithm consumes subsequent frames in order and emits a point candidate, an acceptance decision and a history state. Later sparse target masks score whether the candidate center is inside the annotated target. This metric does

not independently verify object identity at every frame, define first visible return, or reveal a true hidden coordinate.

**What changed.** The original system exposed three spatial hypotheses but never used them to choose a candidate. Its two-frame confirmation and appearance control made identical decisions at all 31 scored Office endpoints. This revision replaces the weak grayscale frontend with frozen learned features and integrates the hypotheses into ranking. That change is implemented and tested; benefit is an empirical question.

**Contribution boundary.** We do not claim that immutable memory, motion models, object permanence, or alternative trajectories are new. The concrete deliverable is a matched, causal experiment separating candidate availability from association quality. Its negative result identifies a necessary next requirement: physical-relation evidence must justify motion transfer instead of treating nearby texture as target motion. That relation-aware extension is not implemented in this release.

## 2. Position relative to prior work

TCOW already addresses target, container and occluder segmentation under permanence assumptions[1]. Object-permanence reasoning predates this project[45, 46], and amodal tracking is itself an established evaluation problem[47]. A humanoid-shaped interface does not change those facts.

XMem and Cutie provide learned video memories[40, 41]. SAMURAI integrates motion-aware memory[42]; DAM4SAM addresses distractor-aware memory[44]; SAM2Long maintains multiple segmentation histories[43]. These are substantially closer competitors than a stationary point. The present DINO association control is an implemented learned baseline, not a reproduction of those tracking systems. Their full matched evaluation remains outstanding.

**Point correspondence and representation** Point tracking, flow, segmentation, generic visual features and detection provide different frontends. Their outputs and supervision are not interchangeable: a query point does not provide the same identity evidence as a full mask. Our matched variants share the same frozen encoder and mask pooling. [2–11]

**Association and temporal identity** Multi-object tracking literature studies association, motion and identity. Its existence rules out claiming that adding a spatial prior to appearance is by itself novel. Bounding-box or multi-target metrics cannot be silently substituted for the sparse point-in-mask metric used here. [12–17]

**Object-centric representation and physical reasoning** Object-centric representations and physical-reasoning datasets motivate richer latent state. They do not establish that our local-flow alternative identifies a container or obeys physical transport. We include these sources as research context, not as executed benchmarks. [18–28]

**Video understanding and observation policies** Video understanding and egocentric datasets broaden the possible evaluation setting. Active-learning theory formalizes information gathering, but a text request for another view is not an executed observation policy. We have not measured the value or cost of a changed camera pose. [29–34]

**Selection and uncertainty** Selection, calibration, conformal prediction and ensembles distinguish score magnitude from empirical coverage. Our fixed gates and Gaussian variance are not calibrated by these methods; referencing their literature does not confer a statistical guarantee. [35–39]

## 3. Executed learned association

### 3.1. Immutable identity and candidate generation

Images are resized to $322 \times 238$, giving a $23 \times 17$ grid of ViT-B/14 patch tokens. We use the official pretrained DINOv2 weights, frozen throughout inference. The normalized mean of tokens selected by the initial query mask gives anchor $a_0$. A mask downsample with area interpolation selects cells above 32 on the original 0–255 scale; if none survive, the query-mask centroid selects one cell. The source mask therefore provides more information than a single point prompt.

At time $t$, each normalized feature $f_{ti}$ receives cosine score $c_{ti} = a_0^\top f_{ti}$. We retain 12 peaks, suppressing a two-token neighborhood after each peak. Candidate centers lie on the patch grid. Every compared variant receives precisely these candidates and similarities. The fixed grid is a limitation for very small targets; a center can miss a target even when the patch overlaps it.

### 3.2. Spatial alternatives affect the decision

Each variant stores last accepted center $p$, velocity $v$, local-flow coordinate $q$, elapsed time $\tau$, and mixture weights $w$. Three conditional means are

$$\mu_1 = p, \quad \mu_2 = p + v\,\min(\tau, 2), \quad \mu_3 = q. \tag{1}$$

The local-flow coordinate follows median Farneback flow in an $11 \times 11$ window around itself. Flow receives only the previous and current frames. Its texture may belong to

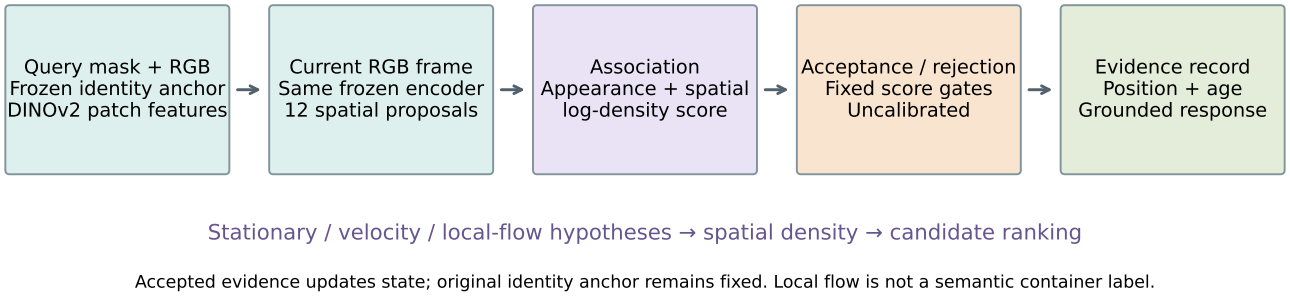



Figure 2. **Implemented information flow.** All variants share the frozen identity anchor and learned candidate frontend. Spatial terms enter association, and accepted candidates update their own histories. The local-flow branch is a geometric alternative, not a learned container relation. Speech is a downstream evidence formatter; a working Tritium speech connection is not established.

the occluder, which is why its interpretation is conditional rather than semantic.

We define isotropic component variance $\sigma^2 = 144 + 400\tau$ in squared pixels and density

$$p_t(x) = \sum_{h=1}^{3} w_h \mathcal{N}(x; \mu_h, \sigma^2 I). \tag{2}$$

The appearance variant uses $s_j = 12c_j$. The motion control adds the log density of component two; the mixture adds

$$s_j = 12c_j + 0.25 \log\big(\max(322 \cdot 238\, p_t(x_j), 10^{-8})\big)\,. \tag{3}$$

The uniform-density ratio makes the logarithm dimensionless. Neither its weight nor the variance was fitted to held-out labels. These are explicit model assumptions, not learned physical probabilities.

### 3.3. Acceptance and state update

The highest-scoring candidate is accepted when its original-anchor cosine is at least $0.65$ and the gap to the next association score is at least $0.15$. An accepted coordinate updates velocity by a half-weighted moving average of displacement divided by elapsed time. It resets the evidence age and local-flow coordinate. Rejected candidates do not replace the last supported point.

After acceptance, hypothesis weights update proportionally to their likelihood of the accepted candidate, with a five-percent uniform floor:

$$w'_h = .95 \frac{w_h \mathcal{N}(x_j; \mu_h, \sigma^2 I)}{\sum_k w_k \mathcal{N}(x_j; \mu_k, \sigma^2 I)} + .05/3. \tag{4}$$

This update is self-reinforcing if an incorrect association is accepted. It is not independent evidence that a physical relation holds. Rejected observations increase dispersion but do not identify whether the target remains present or has left the image.

### 3.4. Controlled appearance adaptation

The adaptive variant retains $a_0$ and an auxiliary anchor $a_t$. Ranking uses $0.8 a_0^\top f + 0.2 a_t^\top f$. Only an accepted candidate with original cosine at least $0.8$ and gap at least $0.5$ may update the auxiliary anchor, through a normalized moving average with rate $0.05$. Candidate generation still uses $a_0$. This bounds one route to forgetting, but can prevent adaptation to appearances absent from the original proposal set. The experiment tests whether this particular rule helps; it is not a claim of optimal memory admission.

**Temporal access.** The encoder processes one frame, flow uses two consecutive frames, and all state depends on earlier outputs. By induction, a prediction at frame $t$ is a function of initialization and frames up to $t$. The evaluator reads endpoint target masks after generating the clip predictions. Five implementation tests check selection influence, rejection behavior, immutable anchoring, prefix determinism and grounded response wording. They do not test scientific superiority.

## 4. Evaluation and version accounting

**Collections.** We execute all eligible clips in two local TCOW collections: 22 Office clips with 31 endpoints and 14 Cup Games clips with 25 endpoints. Cup Games is derived from Perception Test; these are two TCOW collections, not two independent benchmark families. All 7,530 intervening frames are processed. Sparse masks constrain which performance claims are possible, even though every frame has a stored prediction.

**Development exposure.** The Office collection was inspected during earlier iterations; Cup Games also existed in the prior workspace. Neither is represented as an untouched

held-out test. The four-variant protocol and thresholds were recorded before this run's outcomes, and no variant was retuned after observing them. A separate scene/object-instance manifest for a genuinely reserved evaluation remains absent.

**Metrics.** Candidate hits count predicted centers inside a target mask. Accepted hits and accepted misses condition on the algorithm's gate; abstentions are all other endpoints. We show operating curves by sweeping the association-gap threshold with the cosine floor fixed. Those descriptive curves use the same development labels and are not calibration or test-time operating-point guarantees. Continuous identity switches, time from first visible return, hidden-region coverage and downstream task success remain unmeasured.

**Earlier results.** The classical implementation produced nine candidates inside the mask and seven accepted hits on Office, with no accepted misses and 24 abstentions. Its appearance-only decisions were identical at every scored endpoint. A separate expanded-search diagnostic obtained 13 top-ranked hits and top-20 oracle set recall of 17. The new learned protocol changes representation, query pooling, candidate scale and acceptance. Its 22 candidate hits cannot be attributed solely to one variable by comparison with those older numbers; the four learned variants are the matched experiment.

## 5. Results: better proposals do not validate memory

On Office, learned appearance yields 22 candidate hits, 11 accepted hits, one accepted miss and 19 abstentions. The spatial mixture has 19 candidate hits, the same 11 accepted hits, three accepted misses and 17 abstentions. It answers two more endpoints, but these additional answers do not increase the accepted-hit total. Single-motion association and controlled adaptation perform worse in accepted hits as well.

On Cup Games, appearance yields eight candidate hits, six accepted hits, 11 accepted misses and eight abstentions. Each spatial variant yields seven candidate hits, six accepted hits, 12 accepted misses and seven abstentions. The increased coverage again corresponds to more error, not more correct answers. This collection exposes a failure concealed by the earlier prototype's very low coverage: a permissive learned matcher can accept many similar distractors.

The spatial mixture changes 1,934 frame selections relative to appearance, while single motion changes 1,843 and adaptive memory changes 1,949. Thus the mechanism is active, unlike the earlier display-only hypotheses. Activity is not utility. Endpoint outcomes show no supported superiority, and missing intermediate labels prevent estimating how many changed frame selections are correct.

The complete execution, including decoding, frozen feature inference, optical flow and four selectors, took approximately 459 seconds on the local MPS environment. This is a combined wall-clock measurement rather than isolated model throughput or a comparison to published GPU speed. Model and protocol hashes accompany the release.

## 6. Qualitative evidence and failure mechanisms

**A matching frontend remains essential.** A verifier cannot recover an object absent from its candidate set. The classical audit proved this on the finite evaluated examples: even an oracle accepting or rejecting its single proposal could yield at most nine correct answers. A larger classical bank improved proposal availability but did not solve association. Learned features further change availability, while patch-grid resolution and appearance similarity remain limiting factors.

**Spatial confidence can be wrong.** A stationary or velocity hypothesis assumes a motion regime. Long occlusion, hand interaction or a moving container can violate it. A mixture of incorrect motion assumptions does not become informative merely because it has multiple components. Likelihood updates based on a mistaken accepted match can reinforce the wrong component.

**Context motion is not containment.** Local optical flow may follow the covering box instead of the target. In one physical situation, following the box is useful because the object is inside it. In another, the object remains behind it and the same update is harmful. The current model has no visual relation classifier to distinguish these situations. This is the specific missing signal motivating the next research hypothesis, rather than a capability already demonstrated.

**Stored identity versus adapted appearance.** Immutable evidence resists destructive updates but cannot cover arbitrary viewpoint change. Controlled updates can still admit a wrong object with high semantic similarity. The adaptive variant's worse Office result makes that tradeoff visible. No claim about optimal adaptation follows from one fixed update rule.

## 7. Official Ameca integration and its boundary

We used the authenticated Tritium interface to create *Occlusion Memory – Evidence Replay* from the official Ameca Gen 1 template. Native 20-second blinking and breathing

| Collection | Variant | Cand. | Hit | Miss | Abst. |
|---|---|---|---|---|---|
| office | DINO appearance | 22 | 11 | 1 | 19 |
| office | + single motion | 18 | 10 | 3 | 18 |
| office | + spatial mixture | 19 | 11 | 3 | 17 |
| office | + controlled update | 18 | 10 | 4 | 17 |
| cup games | DINO appearance | 8 | 6 | 11 | 8 |
| cup games | + single motion | 7 | 6 | 12 | 7 |
| cup games | + spatial mixture | 7 | 6 | 12 | 7 |
| cup games | + controlled update | 7 | 6 | 12 | 7 |

Table 1. **Completed learned-frontend experiment.** All variants in a collection share the encoder, candidates, initialization mask and causal access. Cand.: in-mask candidate centers; Hit/Miss: accepted in-mask/out-of-mask endpoints. Office has 31 endpoints and Cup Games 25. Neither memory variant improves accepted hits over learned appearance.

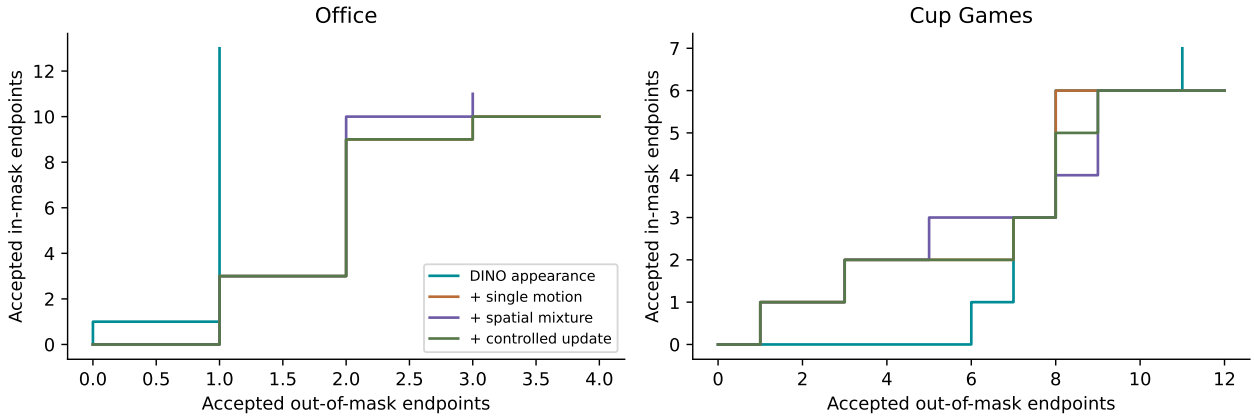



Figure 3. **Usefulness and error together.** Descriptive operating curves count accepted in-mask and out-of-mask endpoints as the association-gap threshold varies. The cosine floor stays at 0.65. Curves are from inspected data, not frozen test thresholds. Coincident segments and tied variants are retained; they should not be interpreted as absent experiments.

tracks were generated and saved. This uses the manufacturer's Animator rig rather than a substitute drawing or a hand-built imitation.

Responsive speech has not been demonstrated. The selected target ad-0005 reports disconnected; the speech editor reports no available voices and asks for appropriate permissions. The Scripts page remains loading, so no robot-side script was installed or executed. These are observed integration limitations, not model results.

The released response generator accepts an evidence record and a caller-supplied, documented speech function. It can describe stored image properties, the last supported image coordinate, elapsed evidence age and unresolved alternatives. It explicitly distinguishes those measurements from unobserved material, mass, hidden surfaces and exact hidden position. A functioning transport to Tritium is still required. Native idle animation is not evidence of perception or transfer.

## 8. What is established, and what remains open

The implemented spatial mechanism now influences candidate selection, a learned frontend is evaluated on two collections, all variants share its information, and both favorable and unfavorable outcomes are released. The original 31-endpoint result is preserved rather than replaced by the later diagnostic. Forty-seven primary-source references position the work relative to existing approaches.

The central research goal is still unresolved: trustworthy identity recovery at useful coverage beyond strong learned trackers. A next claim would require a learned or independently supported relation model; matched comparisons with modern causal tracking systems; full-sequence identity and visibility annotations; calibration separate from evaluation; and independent scene/object groups. Hidden-location and active-observation claims additionally require appropriate ground truth or controllable viewpoints. Those omissions cannot be repaired by a more impressive avatar, a longer bibliography, or labeling this report a 10/10 paper.

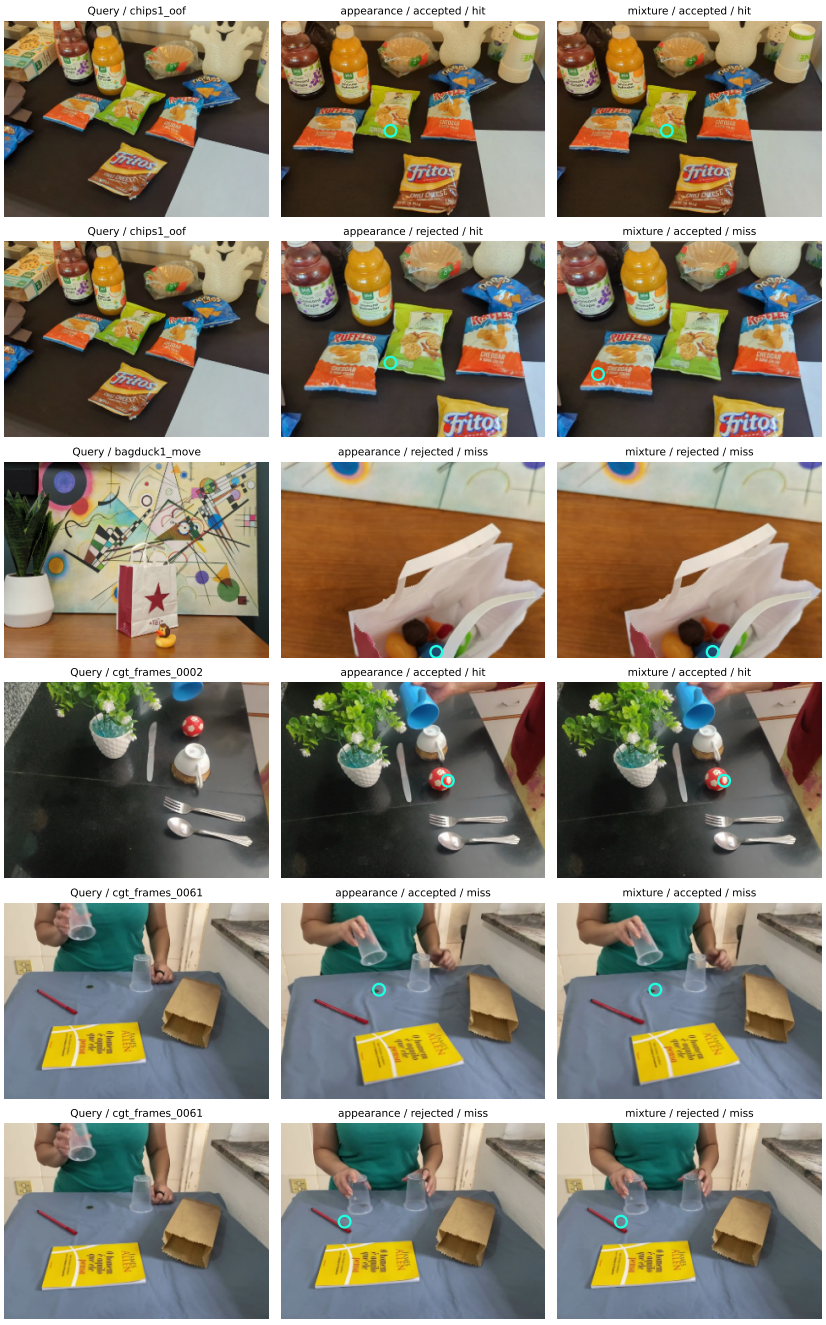



Figure 4. **Successes, accepted errors and abstentions.** For each collection, the gallery shows the lexicographically first spatial-mixture accepted hit, accepted miss and abstention. Left: query frame. Middle/right: the same evaluated frame with appearance/mixture candidates. These outcome-selected illustrations are not prevalence estimates; all endpoint records and the exact selection manifest are released.

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